

HW4: Home Loan Approval Analyzer

About

As houses become increasingly expensive in America, millions are seeking loans to finance their dream homes. However, the approval process can be opaque. My aim is to investigate why some mortgage applications were granted while others were rejected.

Using partial dependence profiles (PDP), I constructed a Shiny visualization that allows prospective homeowners to discern the likelihood of loan approval based on various customer details. This visualization answers:

- How do applicant characteristics and their interactions affect one's chances of obtaining a home loan? Do these influences reveal anything about the fairness/unfairness of the lending process?

My visualization leverages an [extreme gradient boosting](#) model (XGBoost) because of its ability to unearth the nuanced relationship between mortgage approval and each applicant characteristic. The PDP plot then concisely visualizes these dynamics based on the user-selected explanatory variables.

My application also allows switching to the Ceterus Paribus (CP) plot. Users can see how a subsample of individuals' loan approval probability changes, underlaid behind the highlighted average probability. However, the CP plot can be too cluttered, and it is unable to meaningfully visualize categorical customer details including gender and education. For that matter, users have to turn solely to PDP. Nonetheless, this plot also has a limitation. Its inability to simultaneously

visualize categorical and numerical variables restricts a holistic analysis of how different features influence loan approval. The only workaround is to plot these two data types separately.

Beyond visualizing individual features, users can specify a grouping variable and identify interaction trends among customer details. Using our understanding of the real world, we can reason out that the relationship between mortgage approvals and any individual credential is not the same across all demographics. Therefore, this functionality highlights how a potential third metric might influence such relationship.

Key Findings

One relationship that stands out is between mortgage approval and credit history. A customer whose credit history meets the guidelines has a significantly better chance of obtaining a home loan than those with a bad credit history. This makes sense because a well-established credit history serves as a reliable indicator of financial responsibility. Lenders can be assured that these customers will repay their home loan on time.

From another perspective, monthly co-applicant income has a clearer positive influence on securing a home loan than the applicant's monthly income, which exhibits a more complicated relationship with loan approval. My best reasoning is that the effect of monthly applicant income on home loan approval is interdependent with other customer details. This is not the case for the co-applicant, as we only have data about their income.

On its own, gender and education impact home loan approval, with males and university graduates having a higher chance of obtaining a mortgage. Taking a step further and setting these variables as the grouping/interaction term, I notice discernible patterns in mortgage approval that allude to the lending process being unjust. Loan requests filed by females or non-graduates are

generally rejected, and this observation holds true across all features. My understanding is that graduates are more financially literate than non-graduates. Gender-wise, females usually take career breaks for childcare, affecting their capacity to repay debts. Nonetheless, these are not reasons to disfavor female or non-graduate applicants as other criteria may reveal that they are just as qualified as men or graduates in securing home loans.

Visualization Creation Process

Kaggle's [Home Loan Approval](#) dataset required several preparations. I removed outliers in the income variables to ensure a more accurate model. I also imputed data by using random sample for categorical features and mean for numerical variables. Additionally, I performed oversampling to create a balanced loan dataset. Because grid search takes a long time to run, I decided to tune the XGBoost hyperparameters manually according to the best combinations generated by this algorithm.

I designed the interface with the idea of making it easily navigable. I allowed users to visualize the relationships either with a CP plot or a PDP plot, controlled by radio buttons. For both plot types, I added the ability to select a grouping/interaction feature and multiple variables for seamless comparisons between applicant characteristics. Furthermore, I used relevant color encodings to represent each categorical variable for recognizability. Because of the previously mentioned limitations, I included the option to visualize categorical features only for the PDP plot. As an added feature, I developed a simple predictive tool that simulates an actual mortgage application where users can discover the predicted status of their loan requests.